Multiple Choice questions	
1.	What is the basic difference between JavaScript and Java? a) Functions are considered as fields b) Functions are values, and there is no hard distinction between methods and fields c) Variables are specific d) There is no difference
2.	Which statement is used to execute a block of code multiple times in JavaScript? a) for b) if c) return d) break

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3.	Why event handlers are needed in JS? a) Allows JavaScript code to alter the behaviour of windows b) Adds innerHTML page to the code c) Change the server location d) Performs handling of exceptions and occurrences
4.	What is the purpose of JavaScript in web development? a) To structure a web page b) To style web page c) To add interactivity and dynamic content to web pages d) To store data on the server
5.	In JavaScript, which of the following is a valid variable name? a) 2names b) $name c) -name d) @name
6.	Which of the following class is used to create a responsive navigation bar in Bootstrap? A) navbar-fixed B) navbar-expand C) navbar-light D) navbar-collapse
7.	What is the purpose of the card class in Bootstrap? a) To create interactive buttons. b) To apply borders to an element. c) To create a flexible content container with optional headers and footers. d) To align text within a grid.
8.	How can we write comments in CSS? a) /* */ b) // c) # d) All of this
9.	Which CSS property is used to control the transparency of an element? a) Opacity. b) Visibility c) filter d) None of them
10.	Which of the following CSS property is used for controlling the layout? a) display b) color c) text d) None of them
11.	Which of the following components of the CSS box model are transparent? a) padding b) margin c) both a & b d) content
12.	What is the function of the HTML style attribute? a) It is used to add style to an HTML element b) It is used to uniquely identify some specific styles of some element c) Both a & b d) None of the above

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13.	What tag is used to render an image on a webpage? a) img b) src c) image d) none of them
14.	The most basic part of any HTML page is? a) ASCII Text b) Binary Text c) Text d) None of the above
15.	How many characters can be written in 1KB? a) 1048 b) 1024 c) 2048 d) None of them
16.	There are the two main types of layouts available in Bootstrap. Write their names and function.
17.	Write down the Boilerplate of HTML.
18.	Why do we use the CSS box-sizing property?
19.	Write the key difference between margin and padding in CSS.
20.	Explain JSON and Provide an example.

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21.	How do we use JavaScript's timing methods, and why are they important?
22.	Differentiate between DOM and BOM in JavaScript.
23.	Can you explain the differences between an HTML Tag, Element, and Attribute?
24.	What are the different types of HTML elements? Write their name and function.
25.	Define function in JavaScript with example.
26.	What is Array in JavaScript? Give an example.

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27.	Create a JSON Array that contains of four students name, email, mobile & address.
28.	What are the different types of loops (repeated statements) in JavaScript and how do they work?
29.	There are different types of conditional statements in JavaScript. Write, why are they used for?
30.	When working on web design and development projects, which code editor do you choose and why?

Feedback to candidate:			
Assessment decisions for this assessment activity: O **Competent** O **Not Yet Competent**			
Candidate's Signature:		**Date:**	
Assessor's Signature:		**Date:**	

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Answer Web Design and Development

1. What is the basic difference between JavaScript and Java? - B
2. Which statement is used to execute a block of code multiple times in JavaScript? - A
3. Why event handlers is needed in JS? - A
4. What is the purpose of JavaScript in web development - C
5. In JavaScript, which of the following is a valid variable name? - B
6. Which of the following class is used to create a responsive navigation bar in Bootstrap? - B
7. What is the purpose of the card class in Bootstrap? - C
8. How can we write comments in CSS? – A
9. Which CSS property is used to control the transparency of an element? – A
10. Which of the following CSS property is used for controlling the layout? - A
11. Which of the following components of the CSS box model are transparent? - C
12. What is the function of the HTML style attribute? - A
13. What tag is used to render an image on a webpage? - A
14. The most basic part of any HTML page is? - A
15. How many characters can be written in 1KB? - B
16. There are the two main types of layouts available in Bootstrap. Write their names and function.

    The two main types of layouts available in Bootstrap are:

    1. Fixed Layout
    2. Fluid Layout (or Responsive Layout)

    Fluid Layout- This layout is necessary for creating an app that is 100 % wider and covers all the screen width.

    Fixed Layout- It is used only for a standard screen (940px). Both layouts can be used for creating a responsive design.

17. Write down the boilerplate of HTML.

```
<!DOCTYPE html>
<html>
    <head>
        <meta charset="UTF-8"/>
        <title>Document Title</title>
    </head>
    <body>

    </body>
</html>
```

18. Why do we use the CSS box-sizing property?

    Box sizing property remove/alter Border & Padding from height & width property in CSS.

    Or

    The box-sizing property in CSS ensures that padding and borders are included within the total width and height of an element, preventing unexpected size adjustments. It is commonly used to simplify layout design.

19. Write the key difference between margin and padding in CSS. [Any one]

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a. Margin is used to create space around elements and padding is used to create space around elements inside the border.
b. We can set the margin property to auto but we cannot set the padding property to auto.
c. In Margin property we can allow negative or float number but in padding we cannot allow negative values.

20. Explain JSON and Provide an example.

JSON is a standard data interchange format that's used to store simple data structures and objects. It's a lightweight, language-independent format that's easy to learn and troubleshoot

Var student = **{name: "Rubel", city: "Dhaka"}**

21. How do we use JavaScript's timing methods, and why are they important?

There are two timing methods in JavaScript: setInterval() and setTimeout().

These methods are used for **creating task repeatedly**.

setInterval() is used to execute a task repeatedly at specified intervals (in milliseconds) **or** execute a task repeatedly within given time period.

setTimeout() is used to execute a task once after a specified delay (in milliseconds) **or** execute a task one time within given time period.

22. Differentiate between DOM and BOM in JavaScript.

DOM (Document Object Model): DOM is a programming interface for HTML and XML documents.
BOM (Browser Object Model): It allows users to interact with the browser. A browser's initial object is a window. As a result, you may call all of the window's functions directly or by referencing the window.

23. Can you explain the differences between an HTML Tag, Element, and Attribute?

HTML Tag is a syntax that is used to print data & define data print.
HTML Element contain starting tag & ending tag with content.
HTML Attribute provide additional information about HTML Tag

24. There are two main types of HTML elements. Write their name and function.

The two types of HTML Elements are-

Block-level Elements: These elements occupy the **entire width available**, starting on a new line. **Or** hold the whole area. //For example, , <p>, <h1>.

Inline Elements: These elements **only take up as much width as necessary** and do not start on a new line. **Or** hold only contain area. //For example, , <a>, .

25. Define function in JavaScript with example.

A function is a block of code designed to perform a particular job. It does not execute on its own and must be called to run. Functions promote code reuse and modularity.

```
function hello()
{
        Document.write("Hello World.");
```

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```
}
hello(); // This will output: Hello World.
```

26. What is Array in JavaScript? Give an example.

Array is a special variable in JavaScript. Array can store multiple values in a single variable. In JavaScript it is possible to store different type of data in an Array to manage data easier.

```
Var data = [10, 20, "Hello World.", 20.34, true, false];
```

27. Create a JSON Array that contains of four students name, email, mobile & address.

```
Var students = [
    {name: "Rakib", email: "rakib@gmail.com", mobile: "01918171615", address: "Dhaka"},
    {name: "Aakib", email: "aakib@gmail.com", mobile: "01918171616", address: "Dhaka"},
    {name: "Sakib", email: "sakib@gmail.com", mobile: "01918171617", address: "Rajshahi"},
    {name: "Nakib", email: "nakib@gmail.com", mobile: "01918171618", address: "Khulna"},

]
```

28. What are the different types of loops (repeated statements) in JavaScript and how do they work?

In JavaScript there are for, while, do while, forEach, for-In repeated statement.
For & while loops start works when condition is true.
Do-while syntax one time works after check condition, if the condition is true then it works again.
ForEach & For-In loops works for Array.

29. There are four types of conditional statements in JavaScript. Write, why are they used for?

There are four conditional statement syntax in JavaScript. They are if, if-else, if else-if & switch.
if statement: Used to execute a block of code if a condition is true.
if-else statement: Used to execute one block of code if a condition is true and a different block if the condition is false.
else-if statement: Used when there are multiple conditions.
switch statement: The switch statement is used to evaluate a variable or expression against multiple possible values (cases). Each case defines a possible match. When the value matches a case, the corresponding block of code is executed. The break statement is used to exit the switch statement once a match is found and the code is executed.

30. When working on web design and development projects, which code editor do you choose and why?

When working on web design and development projects, many developers prefer using **Visual Studio Code (VS Code)** due to its versatility, wide range of extensions, and strong community support. **VS Code** is widely favored for its speed, powerful features, and flexibility, making it a top choice for web design and development. Student may explain any reason on: User-Friendly Interface, Extensions and Integrations, Built-in Git Support, Cross-Platform, Performance.

Or

Notepad++

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